INCH-POUND

MIL-DTL-17/127D
19 June 2015
SUPERSEDING
MIL-C-17/127C
11 May 1981

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/127-RG393

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

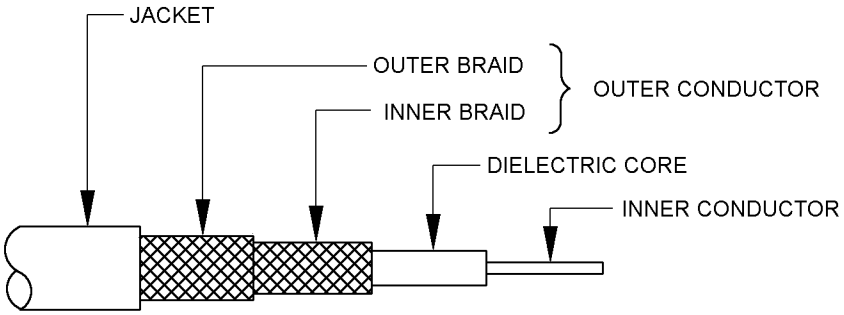


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of silver-coated copper wire, each strand .0312 inch diameter. Overall diameter: .094 inch ± .001.
Dielectric core	Type F-1: Solid, extruded PTFE. Diameter: .285 inch ± .005.
Outer conductor	Double braid of AWG size 34 silver-coated copper wire. Diameter: .360 inch maximum.
Inner braid	Coverage: 95.7% nominal Carriers: 24 Ends: 6 Picks/inch: 16.6 ± 10%
Outer braid	Coverage: 98.0% nominal Carriers: 24 Ends: 7 Picks/inch: 15.4 ± 10%
Jacket	Type IX: FEP. Diameter: .390 inch ± .010.

AMSC N/A

FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 1,875 V rms, maximum.
Operating frequency: 11 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55°C to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 0.152 ohm per 100 feet.
Elongation: 20 percent, minimum.
Tensile strength: Not applicable.

Engineering notes: This cable is useful in general purpose, high temperature applications (see connector series “N” per MIL-PRF-39012).

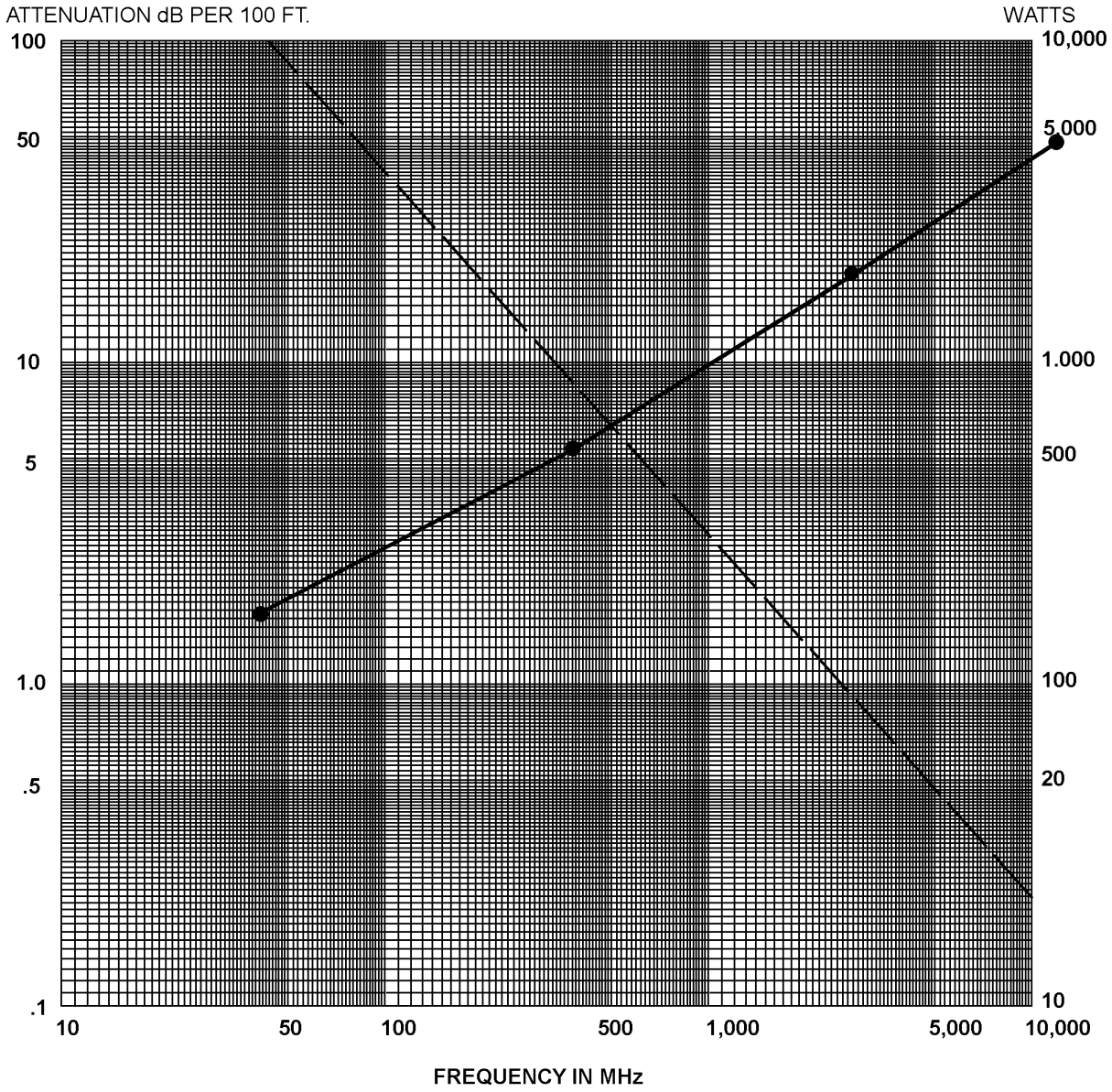
REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.
Environmental and mechanical:
Eccentricity: 10 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 10 pounds, minimum; 50 pounds, maximum.
Aging stability: Not applicable.
Stress crack resistance: +230° ± 5°C; mandrel size 5 times the jacket diameter.
Outer conductor integrity: Not applicable.
Cold bend: -55° ± 2°C.
Dimensional stability: +200° ± 5°C.
Inner conductor from core: .250 inch, maximum.
Inner conductor from jacket: .475 inch, maximum.
Bendability: Not applicable.
Flammability: Applicable.
Weight: 17.5 pounds per 100 feet, maximum.

Electrical:

Spark test: 2,000 Vrms, +10 percent, -0 percent.
Voltage withstanding: 7,500 Vrms, +10 percent, -0 percent.
Insulation resistance: Not applicable.
Corona extinction voltage: 2,500 Vrms, minimum.
Characteristic impedance: 50 ± 2 ohms.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 32 pF per foot, maximum.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.
Contamination: Not applicable.

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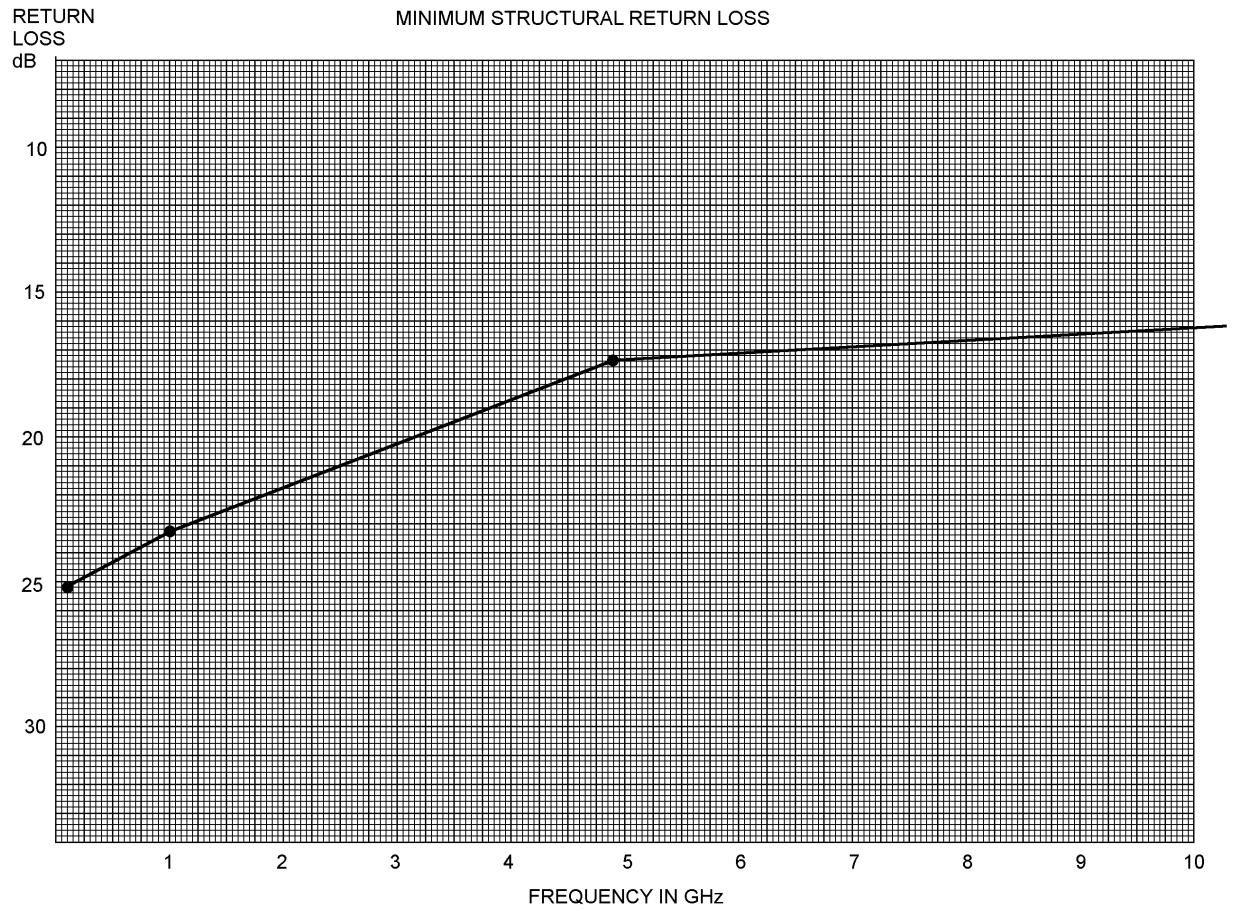
Maximum attenuation ——————

Maximum power at 25°C, sea level — — — —

Attenuation	
MHz	dB
50	1.7
400	5.0
3000	18
11000	45

FIGURE 2. Power rating and attenuation.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

Structural return loss	
MHz	dB
100	25
1000	23
3000	20
5000	17
11000	16

FIGURE 3. Structural return loss.

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Part or Identifying Number (PIN): M17/127-RG393.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2015-020)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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